PROPOSAL · INTERNATIONAL COMPUTE TREATY

# MIRI agreement

A US–China-led treaty would block training runs above a very high compute threshold while controlling smaller frontier work and the hardware that could support a breakout.

## Background and provenance

**WHY IT WAS WRITTEN**

The authors argue that premature artificial superintelligence could create catastrophic loss-of-control, misuse, and geopolitical risks. The introduction starts from the question of what states should do if current development methods cannot safely reach ASI. Instead of calling only for restraint in general terms, the paper supplies a treaty text, operational thresholds, and staged implementation ideas to make a halt more concrete while preserving existing AI applications.

**PEOPLE AND ORGANISATIONS**

Aaron Scher, David Abecassis, Peter Barnett, and Brian Abeyta are identified in the manuscript with the Machine Intelligence Research Institute (MIRI) Technical Governance Team. Their institutional role matters because the paper is a technical-governance design from an AI-safety research organisation. It is not a negotiated agreement, an official US–China position, or a treaty endorsed by the proposed verification authority.

**KEY DATES**

First posted to arXiv on 13 November 2025; version 2 followed on 17 November 2025, and the current version 3 is dated 8 May 2026. This brief uses version 3. The later revision date should not be mistaken for the proposal's first publication.

**EARLIER FOUNDATIONS**

The document draws on prior writing about verifying international AI agreements and on the team's earlier work on chip tracking and compute governance. Its own annex develops treaty articles, historical precedents, cluster thresholds, and a staged search for existing chips. Those sections form one expanded 2025–26 proposal; the verification literature it cites is older than the treaty draft itself.

Sources: Current paper ↗ Version history ↗

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| **AUTHORS** | Aaron Scher, David Abecassis, Peter Barnett, and Brian Abeyta |
| **ORGANISATION / AFFILIATION** | Machine Intelligence Research Institute, Technical Governance Team |
| **ORIGINAL DOCUMENT** | **An International Agreement to Prevent the Premature Creation of Artificial Superintelligence ↗** |

**PURPOSE**

The proposal aims to prevent premature artificial superintelligence and associated extinction or loss-of-control risk. It also cites misuse and geopolitical instability from an ungoverned race. Labour disruption and ordinary market concentration are not its primary rationale.

**WHAT IS PACED**

Training above $10^{24}$ FLOP would be prohibited; runs above $10^{22}$ FLOP would require authorization. Covered clusters, accelerator production and transfers, and dangerous capability-enhancing research would be regulated. Existing permitted inference could continue.

**WHEN AND FOR HOW LONG**

The authors urge negotiation before compute becomes more dispersed and AI progress makes an agreement harder. Their annex sketches staged cluster coverage, beginning with the largest facilities and expanding to smaller ones over roughly two years; the prohibition itself has no preset end date.

**HOW PACING WORKS**

Treaty members prohibit unauthorized large runs, register and monitor covered compute, constrain specified research and chip transfers, and investigate suspected violations. Thresholds and exceptions are managed through the treaty body.

**VERIFICATION METHODS**

A global chip and facility census, production and transfer records, on-site inspections, power and thermal checks, customs and intelligence reporting, and potentially hardware monitoring are layered to detect undeclared clusters or runs. These are proposed tools, not all a single mandatory technical design.

**WHO RUNS IT**

A coalition led by the United States and China establishes an international agreement with an executive council and verification authority. States enforce domestic controls and provide access for treaty inspections.

**LIMITATIONS**

The paper acknowledges the political difficulty of major-power agreement, intrusive access to sensitive facilities, and uncertainty about hidden compute. Algorithmic efficiency gains or new paradigms can weaken a FLOP threshold unless rules are revised.

## Mapped verification methods

10 mechanisms in the atlas map to this text. The labels indicate the strength of each match, not whether the method works.

**Chip production and transfer ledger**
Core

**Physical chip inventory inspection**
Core

**Facility portal and camera monitoring**
Specified option

**Facility power and cooling reconciliation**
Specified option

**Interconnect bandwidth cap**
Specified option

**Cloud workload metering**
Partial match

**Independent predeployment capability evaluation**
Partial match

**On-chip operation metering**
Partial match

**Remote chip location attestation**
Partial match

**Training transcript and checkpoint replay**
Partial match

## Explore the comparison

**See all eight proposals by mechanism →**

This is an analytical summary of the linked proposal. Examples and scenario dates are identified as such; see the original document for the authors' full arguments.